

Quantifying Collagen Extracellular Matrix Organization in the Laryngeal Tumor Microenvironment

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Background

Collagen in the Extracellular Matrix and Spatial Biology

- The tumor microenvironment is shaped by both its cellular constituents and the surrounding extracellular matrix (ECM).
- IMC studies of tumors have focused on cell phenotyping; the structural organization of collagen - the primary ECM component - remains largely unexamined.
- Collagen remodeling and alignment (tumor-associated collagen signatures) accompany tumor progression and invasion.
- We developed a Python (scikit-image) pipeline to extract quantitative descriptors of collagen architecture from the indium-113 collagen channel of 150 IMC-imaged tumor and paired-normal cores from 53 larynx-cancer patients.
- The pipeline resolves the collagen fiber network and yields stable per-core feature vectors, establishing ECM architecture as a quantifiable feature of the tumor microenvironment.

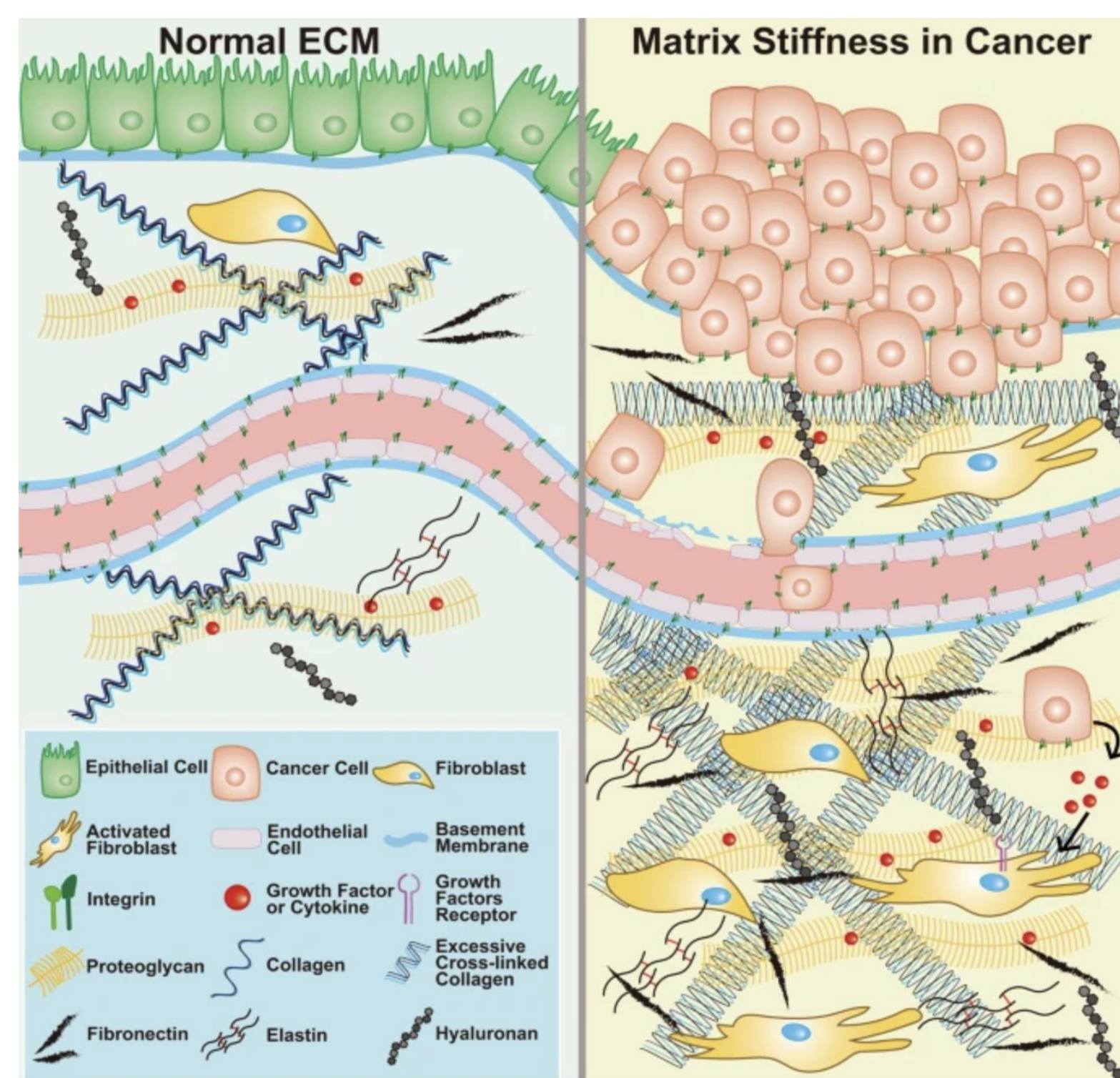


Fig 1. ECM organization in normal tissue (left) versus the tumor microenvironment (right), where excess collagen and cross-linking stiffen the matrix.¹

Data & Imaging

- Imaging mass cytometry (IMC): metal-tagged antibodies bind ~40 proteins in the tissue microarray; a laser ionizes the metals to map each protein at ~1 μm^2 resolution.
- Cohort: 150 tumor and paired-normal cores from 53 larynx-cancer patients. We analyze the indium-113 (In113) collagen channel - 1200x1200-pixel ion-count images.

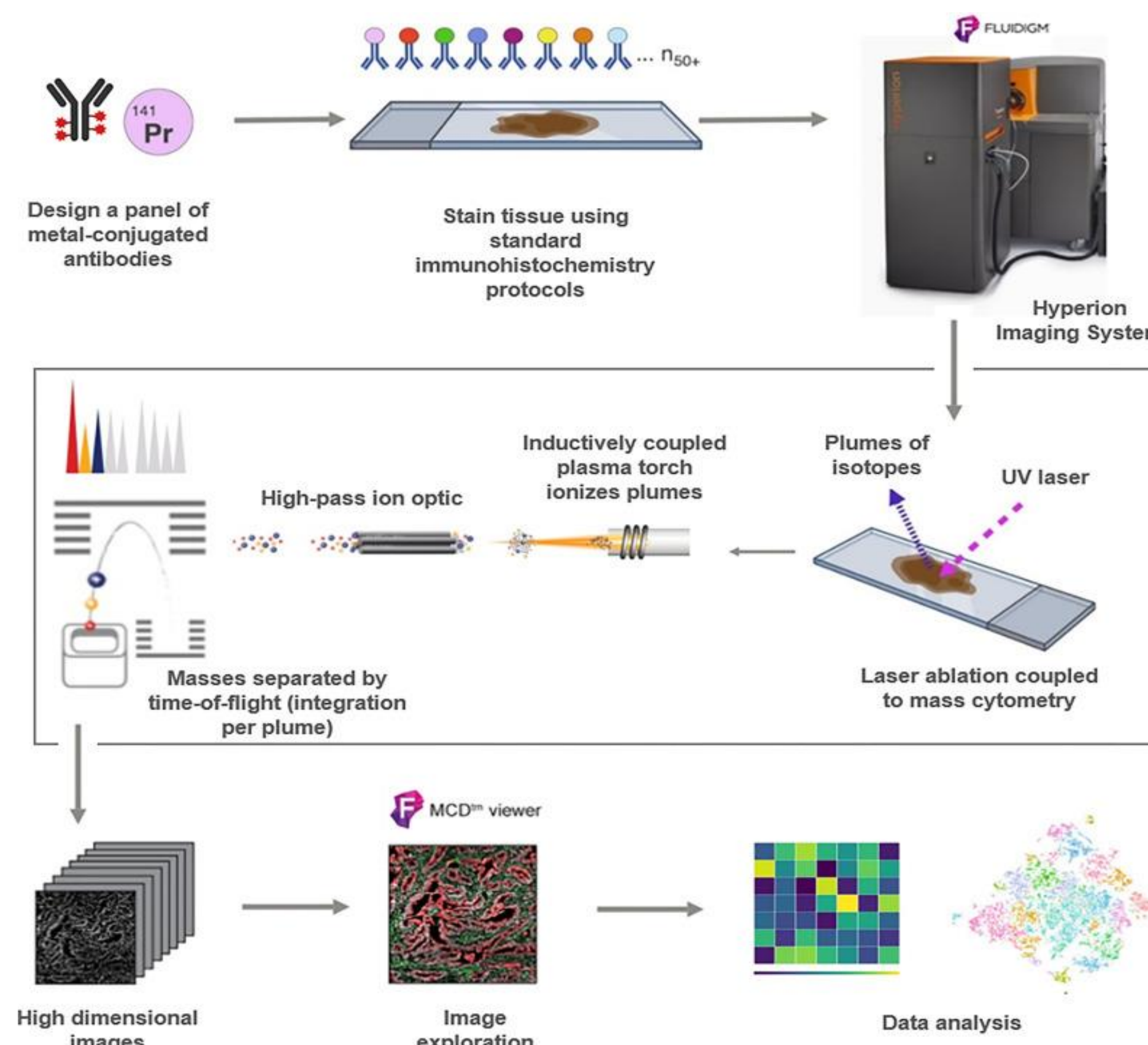
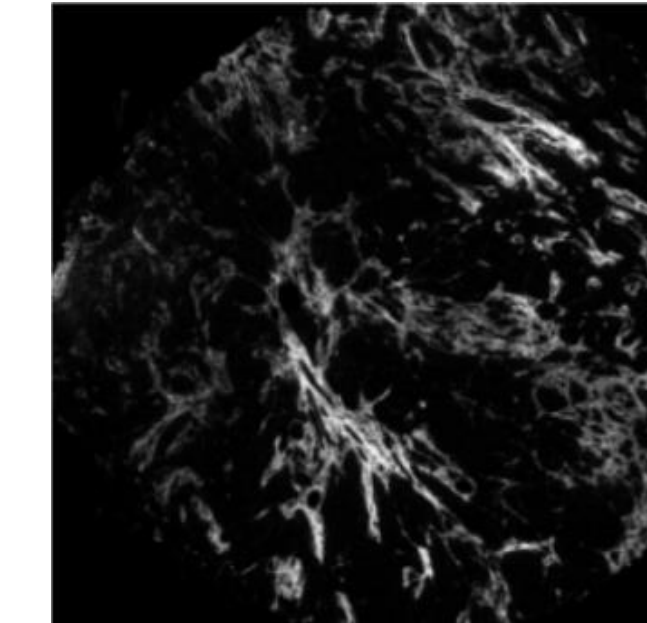


Fig 2. IMC workflow: antibody staining, laser ablation, and time-of-flight mass spectrometry yield multiplexed, spatially-resolved protein images.²

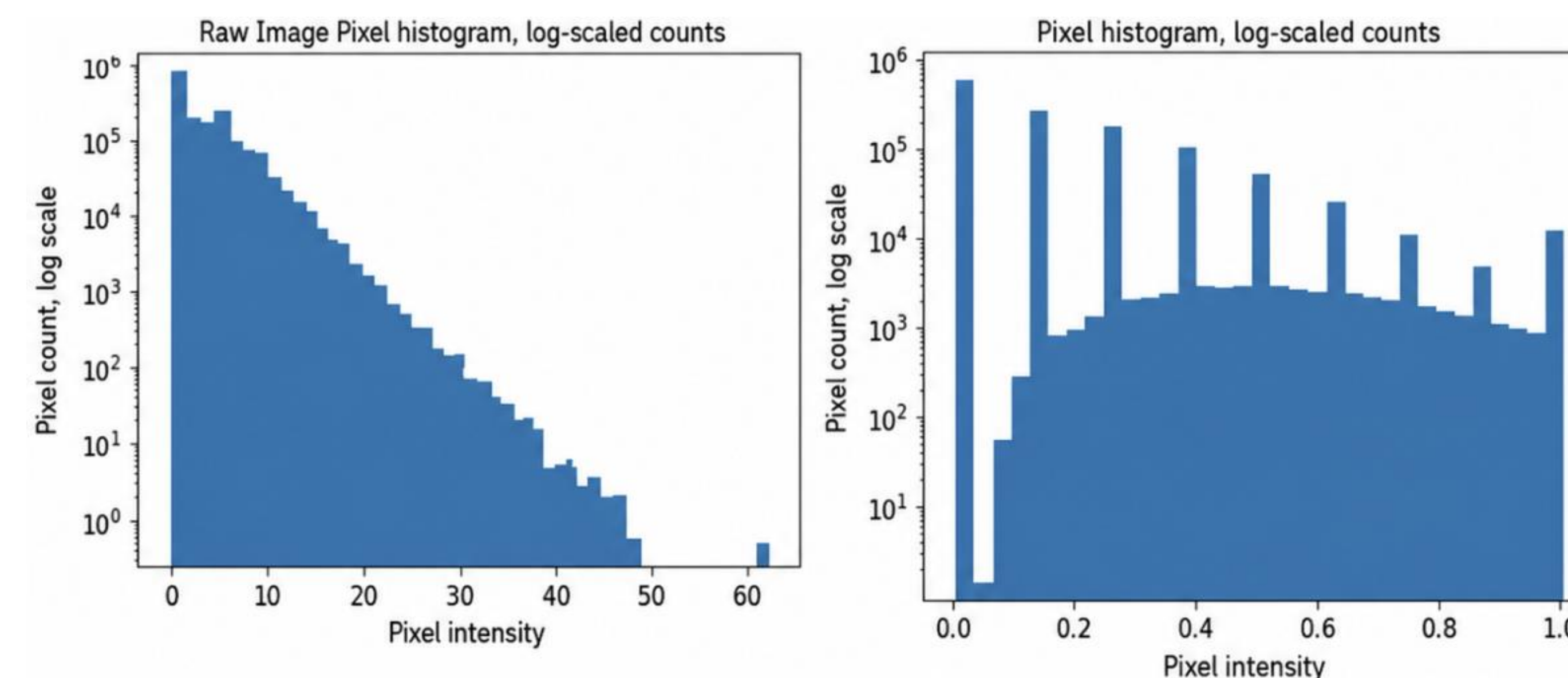
Methods

1. Raw Collagen Input

- Collagen channel of one In113 TMA core (In113_113In_Collagen), read from .tiff
- Pixel value = collagen ion count per spot; signal sparse, mostly low integer counts

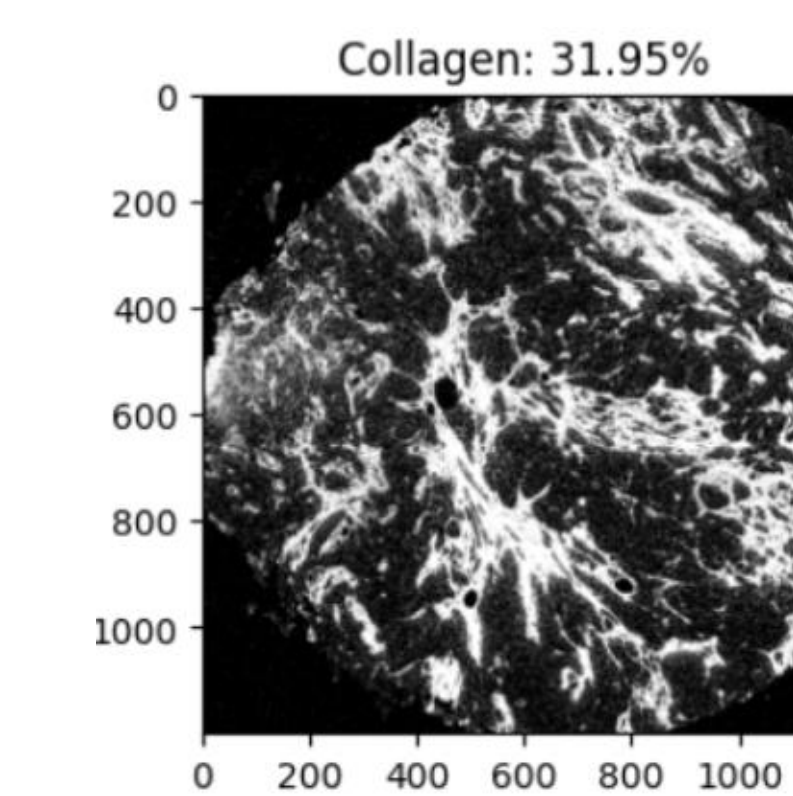


2. Preprocess



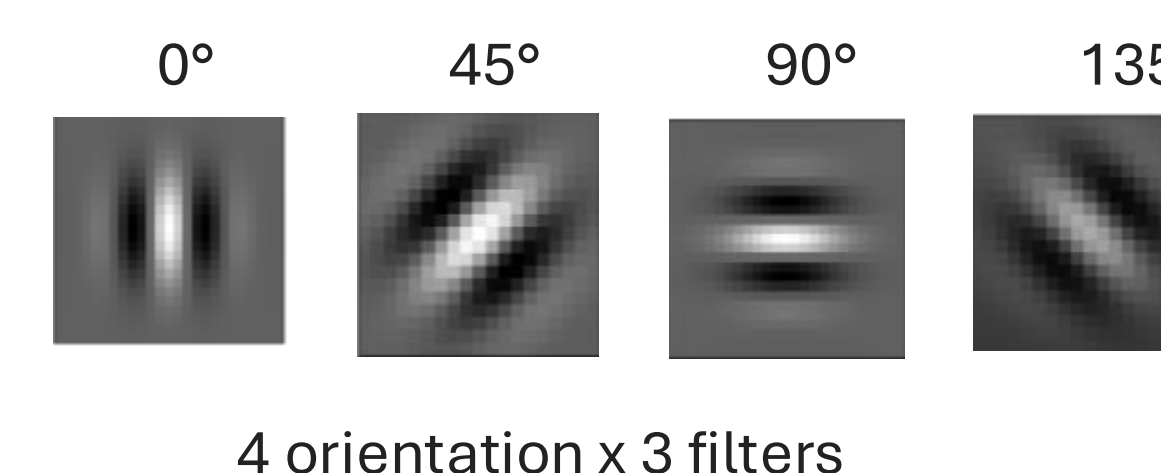
3. Segment Collagen

- Triangle thresholding automatically splits collagen from background into a binary mask
- Area fraction = mean of the mask $\rightarrow$ collagen covers 31.95% of the core



4. Gabor Texture Filters

- Gabor filter bank at 4 orientations (Applied to the collagen to capture directional texture energy as feature columns)



5. Ridge + Skeleton (Fibers)

- Ridge filter enhances tubular fiber structures
- Skeletonize to 1-pixel centerlines $\rightarrow$ fiber-level metrics (density, length) per core

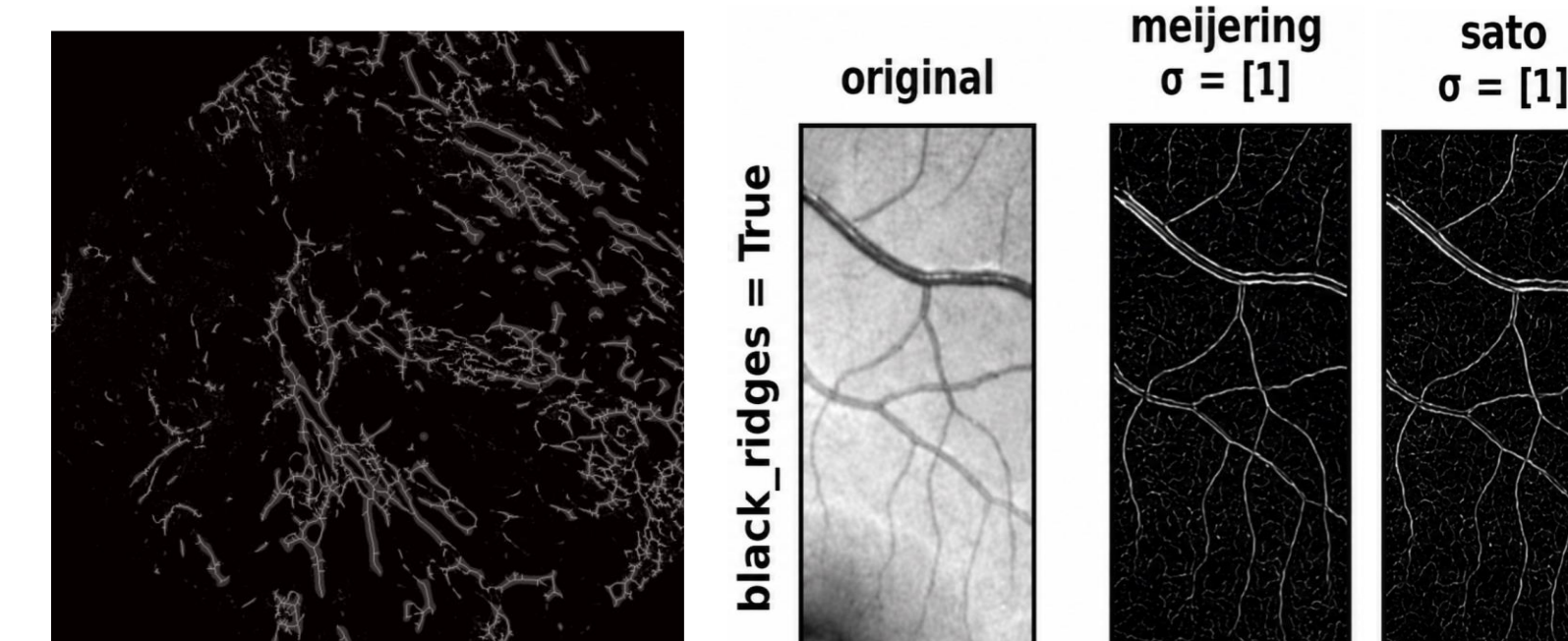


Fig 3. Analysis pipeline applied to one In113 collagen core. Each panel is a labeled step; features are computed per step. Each core becomes a vector of structural collagen features (intensity, texture, fiber-network, alignment).³

Results

Feature Extraction & Correlation

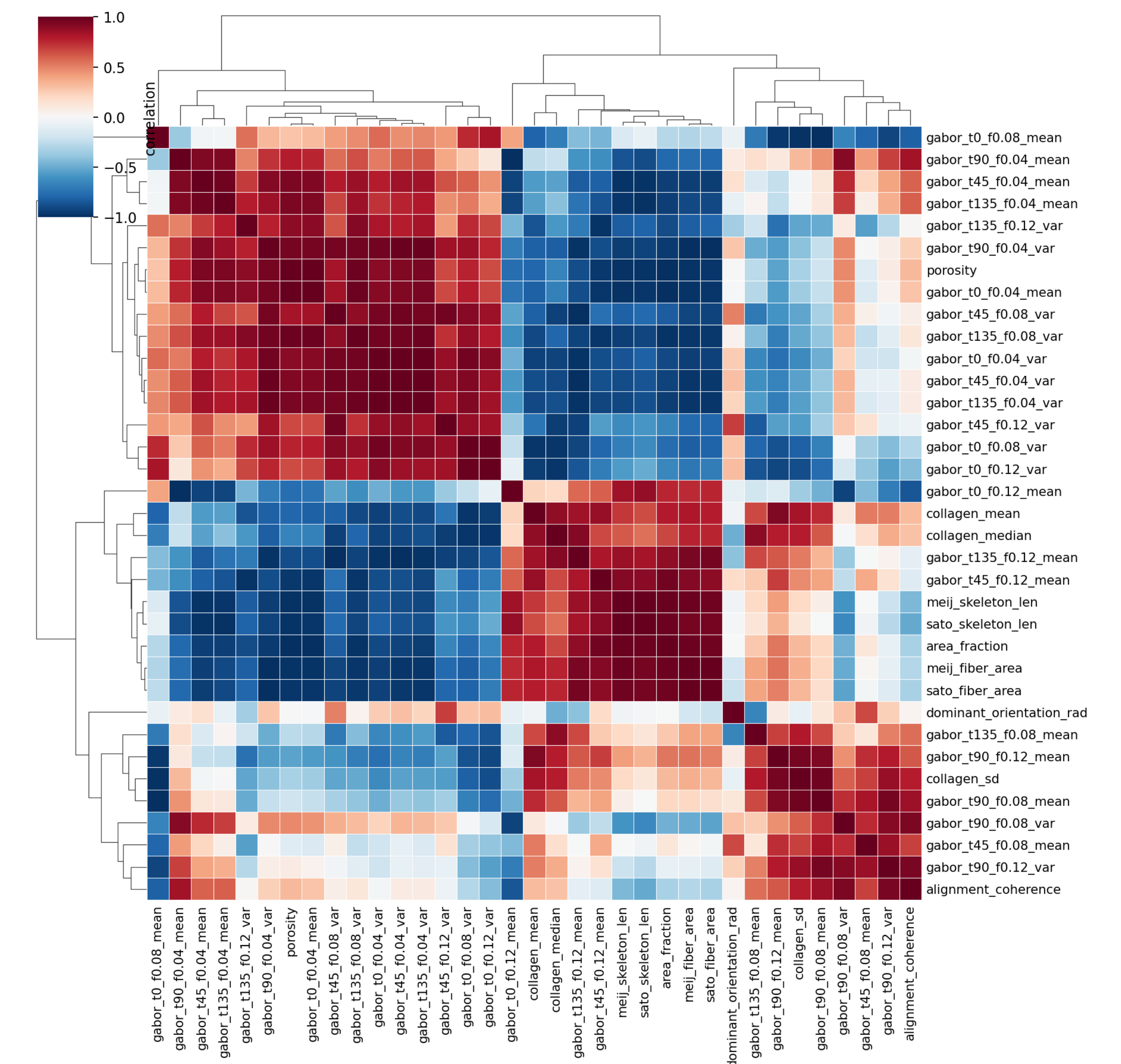


Fig 4. Clustered feature correlation (preliminary). Side bars mark feature families; blocks show which families carry shared information. Texture and fiber features carry overlapping information and cluster into a few families. Keeping one representative per family yields a compact, non-redundant set - the basis for a stable, sparse model.

Discussion

Summary & Future Directions

- The correlation heatmap resolves into interpretable families: collagen density and fiber-network features move together (amount + structure); Gabor texture features form their own blocks across scales; and fiber alignment (coherence) is largely independent.
- So the ~35 raw measurements compress to a handful of non-redundant axes - density, texture, and alignment.
- Next: apply a sparse, stable dimension reduction to select representative features, then test whether they predict survival outcome in tumor cores.

References

- Huang et al., 2021, <https://doi.org/10.1038/s41392-021-00544-0>
- Le Rochais et al., 2022, <https://doi.org/10.3389/fimmu.2022.859414>
- Van der Walt S. et al., 2014, <https://doi.org/10.7717/peerj.453>